

# MuseTune: AI-Driven Songwriting using Sentiment-Based Chord Mapping with a Hybrid LLM and Music Database Retrieval Framework

Arpit Soni

School of Computer Science  
Adelaide University  
Adelaide, South Australia, Australia  
a1897431@adelaide.edu.au

**Abstract**—Songwriting is a multifaceted creative task that requires the careful alignment of lyrical sentiment with harmonic structure, presenting a substantial barrier for aspiring musicians, hobbyists, and even experienced composers exploring new emotional themes. Existing artificial intelligence (AI) tools for songwriting typically address either lyric generation or chord suggestion in isolation, with few integrating both under a unified emotional framework that can be measured rigorously against domain appropriate metrics. This paper presents MuseTune, a fully automated end to end pipeline that converts free text journal entries into 4-line songs accompanied by matching guitar chords and AI synthesized audio. MuseTune combines five modular components: a RoBERTa classifier fine tuned on the GoEmotions dataset for sentiment detection across 27 emotional classes; the proposed ARIA (Affect-driven Retrieval for Intelligent Accompaniment) algorithm that queries a curated 7,200 progression corpus sourced from HookTheory and the Chord Corpus (ChoCo); three state of the art large language models (OpenAI ChatGPT, Google Gemini, and Meta LLaMA) for lyric generation; and an audio synthesis stage that produces a singable WAV file using a Karplus-Strong guitar synthesizer combined with Google Text-to-Speech vocal narration. We evaluate the system through a  $3 \times 3$  factorial experiment across ten emotion categories ( $n = 90$  experiments) using six peer reviewed metrics: harmonic validity computed via music21, chord similarity via MIREX, Spiral Array tonal tension, GPT-2 perplexity for lyric fluency, METEOR for semantic alignment, and re-detected GoEmotions emotional alignment. The Gemini  $\times$  ARIA configuration achieved the highest overall composite score of 0.531, with all three ARIA combinations occupying the top three positions, while ChatGPT  $\times$  LLM-Direct produced the lowest score at 0.438. The results demonstrate that hybrid LLM and music database approaches significantly outperform single component solutions across all measured dimensions, and that the choice of chord generation method affects the final output quality more strongly than the choice of LLM.

**Index Terms**—AI songwriting, sentiment analysis, chord generation, large language models, music information retrieval, emotional alignment, GoEmotions, ARIA, HookTheory, ChoCo

## I. INTRODUCTION

### A. Background and Context

Artificial Intelligence (AI) has progressed beyond traditional computational domains and entered the realm of human creativity. From visual arts and poetry generation to symbolic music composition and audio synthesis, generative AI systems

now produce outputs once thought to require uniquely human cognition [1], [2]. The intersection of AI and music has emerged as a particularly active research area, encompassing tasks such as melody generation, chord prediction, lyric writing, genre classification, and even full song synthesis through systems like Suno, MusicGen, and AudioLDM.

Among these creative AI tasks, songwriting presents a uniquely complex challenge because it requires the simultaneous coordination of multiple dimensions: lyrical meaning, emotional tone, harmonic structure, rhythmic patterns, and rhyme. Unlike pure text generation or pure audio synthesis, songwriting fundamentally demands cross modal coherence: the words must feel consistent with the harmony, and the harmony must support the emotional intent of the words. A song with sad lyrics over a major key bright progression creates a jarring disconnect that listeners immediately notice.

While modern large language models (LLMs) such as GPT-4, Gemini, and LLaMA can produce remarkably fluent text and even passable poetry [3], and while dedicated music systems can generate chord progressions or melodies independently [4], very few tools combine both lyric and chord generation within a unified, emotionally coherent framework [5], [6]. This integration gap is the central problem MuseTune addresses.

Recent advances in transformer based emotion classification, most notably the GoEmotions taxonomy developed by Google Research [7], have enabled fine grained emotion recognition across 27 distinct categories, far beyond the binary or six basic emotion schemes that dominated earlier work. Parallel to this, the music information retrieval (MIR) community has developed rich annotated datasets such as HookTheory (which links chord progressions to Spotify audio features) and the Chord Corpus (ChoCo) [8] (which aggregates over 20,000 annotated progressions in the JAMS format). These resources, when combined with modern LLMs and proper evaluation methodologies, make it increasingly feasible to bridge the gap between linguistic sentiment and musical harmony in a measurable, reproducible way.

## B. Motivation

Despite the rapid progress of generative AI in music, current systems suffer from three critical limitations that motivate the design of MuseTune.

*First, fragmented pipelines.* Existing tools typically treat lyrics and chords as independent outputs rather than as emotionally interlocked components of a single song. A user wanting to compose a song about grief might use one tool to generate sad lyrics and another to find a minor key progression, with no system level guarantee that the two outputs harmonize emotionally or musically. Even commercial products such as Suno and Udio, while impressive in audio quality, operate as closed black boxes that provide no insight into the emotion to music mapping logic.

*Second, the music theory barrier.* For a novice musician, translating an abstract feeling such as grief, hope, or nostalgic longing into both meaningful words and harmonically suitable chords requires formal music theory knowledge. Concepts such as the circle of fifths, secondary dominants, voice leading, and modal interchange are not intuitive to non musicians, yet they directly determine whether a chord progression supports or undermines a lyrical emotion. This creates a substantial entry barrier that AI tools could in principle remove, but most current tools simply offload the burden to the user by requesting genre or style descriptors rather than emotional ones [5], [9].

*Third, weak evaluation rigor.* Prior work in AI songwriting rarely evaluates outputs through statistically robust experimental designs or domain specific citable metrics. Many papers rely on small qualitative demonstrations (often  $n = 1$  to  $n = 3$ ) or generic NLP scores such as BLEU and ROUGE [10], which were designed for machine translation rather than creative text. These metrics fail to capture musical alignment, harmonic correctness, or emotional coherence. Without proper evaluation frameworks, the field cannot reliably compare approaches or measure progress.

These three limitations together suggest a clear research need: an emotionally aware, fully automated songwriting system whose performance can be measured rigorously against published, domain appropriate metrics, and whose design choices are transparent and defensible.

## C. Problem Statement

The central problem this paper addresses can be stated as follows. Given a free text journal entry expressing an emotional state, generate a 4-line song with chord progression and audio output that (1) correctly identifies the underlying emotion, (2) selects chord progressions harmonically consistent with that emotion and stylistically similar to human composed music, (3) produces lyrics that are linguistically fluent, semantically appropriate, and emotionally aligned with the target, and (4) renders a complete audio artifact that combines these elements into a listenable song. Crucially, the system’s performance must be quantifiable through reproducible, citable metrics rather than subjective judgment alone.

## D. Research Questions

This work investigates three primary research questions:

**RQ1:** *Can an emotion to music retrieval algorithm grounded in a curated chord corpus outperform both pure LLM based and pure music theoretic baselines for emotion conditioned chord generation?*

**RQ2:** *Do different state of the art LLMs exhibit complementary strengths and weaknesses when used for emotion conditioned lyric generation, and if so, what are these specializations?*

**RQ3:** *Can a fully automated evaluation framework based exclusively on peer reviewed metrics produce statistically robust rankings of AI songwriting pipeline configurations across diverse emotional inputs?*

## E. Proposed Solution

This paper introduces **MuseTune**, a fully automated AI songwriting pipeline that converts free text journal entries into 4-line songs accompanied by matching guitar chords and AI synthesized audio. MuseTune integrates five sequential stages: (1) a fine tuned RoBERTa GoEmotions classifier for fine grained sentiment detection across 27 emotion categories; (2) a chord retrieval layer called **ARIA** (Affect-driven Retrieval for Intelligent Accompaniment) that queries a curated database (ARIA-DB) of 7,200 chord progressions sourced from HookTheory and ChoCo; (3) a lyric generation module supporting three state of the art LLMs (OpenAI ChatGPT gpt-4o-mini, Google Gemini 2.5 Flash, and Meta LLaMA 3.3 70B via the Groq inference API); (4) an audio synthesis stage producing a singable WAV file using a Karplus-Strong guitar synthesizer combined with gTTS based vocal narration; and (5) an evaluation engine that scores every generated song across six metrics drawn from the music information retrieval and natural language processing literature.

The system is evaluated through a  $3 \times 3$  factorial experiment (3 LLMs  $\times$  3 chord generation methods) conducted across ten distinct emotion categories, yielding  $n = 90$  total experiments. This design supports rigorous statistical comparison of all nine pipeline configurations.

## F. Contributions

The primary contributions of this work are as follows:

- **Design.** A novel hybrid architecture that combines large language models with music database retrieval to generate emotionally aligned lyrics and chord progressions within a single end to end pipeline, demonstrating that targeted retrieval outperforms pure LLM generation for symbolic music tasks.
- **ARIA Algorithm.** A new chord retrieval algorithm that maps fine grained emotions to chord progressions through a two stage process: a learned GoEmotions to musical bucket mapping followed by valence, energy, and mode filtering over a curated corpus.
- **ARIA-DB Corpus.** A merged 7,200 entry chord progression database aggregating HookTheory (pop and rock progressions with Spotify audio features) and ChoCo

(jazz, classical, and folk progressions parsed from JAMS format), with all entries normalized to Roman numeral notation.

- **Implementation.** A fully working modular Python pipeline that produces detected emotion, chord progression, four line lyrics, and a synthesized audio file from any journal input.
- **Empirical Evaluation.** A statistically robust  $3 \times 3$  factorial study ( $n = 90$ ) comparing three LLMs against three chord generation methods using six citable metrics, with results published as a public CSV for reproducibility.
- **Empirical Findings.** Identification of the optimal configuration (Gemini  $\times$  ARIA at composite score 0.531), demonstration that all three ARIA combinations occupy the top three positions in the ranking, and discovery that the optimal chord method varies by emotion category.
- **Reproducibility Package.** A publicly released Colab notebook with reproducible cells, the merged ARIA-DB CSV, the raw experimental results, and six publication ready figures, enabling full replication of all reported findings.

## G. Paper Organization

The remainder of this paper is organized as follows. Section II provides background on music theory, emotion models, and LLMs for creative tasks. Section III reviews related work in lyric generation, chord generation, emotion modeling, and evaluation. Section IV details the MuseTune architecture and the ARIA algorithm. Section V presents the experimental methodology and results. Section VI discusses findings and implications. Section VII addresses threats to validity. Section VIII concludes and outlines future work. Sections IX and X address ethics and cybersecurity. Section XI describes the replication package.

## II. BACKGROUND AND PRELIMINARIES

### A. Music Theory Fundamentals

At its core, music theory provides a vocabulary for describing how pitches combine in time. A *chord* is a set of three or more pitches sounded together. A *chord progression* is a sequence of chords moving through time, typically defined relative to a tonal center or *key*. The *Roman numeral notation* system describes chords by their function within a key, independent of absolute pitch: in C major, the I chord is C major, the V chord is G major, the vi chord is A minor, and so on. Lowercase numerals (i, ii, iii) denote minor chords while uppercase numerals (I, II, III) denote major chords. This notation enables a single progression such as I-V-vi-IV to be transposed to any key while preserving its emotional character.

The relationship between chord progressions and emotion is well documented in musicology. Major keys are generally perceived as bright, happy, or triumphant, while minor keys evoke sadness, longing, or solemnity. Specific progressions carry strong emotional associations: the I-V-vi-IV progression in major is ubiquitous in optimistic pop music, while vi-IV-I-V

conveys reflective melancholy. The Andalusian cadence (i-VII-VI-V) evokes Spanish or Mediterranean atmosphere, and the i-iv-V-i progression is the canonical minor authentic cadence used in classical and folk music.

### B. Emotion Models in Music and Text

Emotion modeling in computational systems typically falls into two camps. *Categorical models* (such as Ekman’s six basic emotions or the GoEmotions 27 class taxonomy) treat emotions as discrete labels. *Dimensional models* (such as Russell’s circumplex of valence and arousal) represent emotions as continuous coordinates in a low dimensional space. MuseTune adopts a hybrid approach: GoEmotions provides the fine grained categorical input from the journal entry, while valence and energy from Spotify audio features (continuous dimensions) guide the chord retrieval. This combination preserves the interpretability of categorical labels while leveraging the smoother similarity structure of continuous space.

### C. Large Language Models for Creative Tasks

LLMs such as GPT-4, Gemini, and LLaMA are pretrained on web scale text corpora and demonstrate emergent capabilities in creative writing, including poetry, dialogue, and story generation. For songwriting, three properties of LLMs are particularly relevant: (1) familiarity with rhyme and meter from poetry training data, (2) emotional vocabulary breadth across registers and styles, and (3) controllability through prompt engineering. However, LLMs trained primarily on text struggle with symbolic music. They have limited exposure to chord progression annotations, and their outputs in this domain often deviate from established human practice, as demonstrated in our experimental results.

## III. LITERATURE REVIEW

We organize related work into six areas relevant to MuseTune: emotion classification in text and lyrics, AI driven lyric generation, chord generation and harmony modeling, multi-modal creative AI, evaluation methodologies, and datasets. We close with a comparative table positioning MuseTune against prior systems.

### A. Emotion Classification in Text and Lyrics

Sentiment and emotion analysis form the entry point of MuseTune’s pipeline, transforming free text input into a discrete emotional signal that drives downstream chord and lyric selection. Medhat et al. [11] provide a foundational survey of sentiment analysis methods spanning lexicon based, classical machine learning, and deep learning approaches. More recent work has shifted decisively toward transformer based fine grained emotion taxonomies. Demszky et al. [7] introduce GoEmotions, a dataset of 58,000 Reddit comments annotated with 27 distinct emotion categories, which has become a standard benchmark for multi class emotion classifiers. Models fine tuned on GoEmotions, such as the RoBERTa variant adopted in MuseTune, outperform binary positive or negative classifiers by capturing nuanced states such as grief, nervousness, and admiration that are central to songwriting expression.

Ara and Rekha [12] specifically apply LSTM based architectures to emotion classification in song lyrics, demonstrating that sequence models can recover emotional progressions through verses and choruses. Miah et al. [13] show that ensemble transformer methods generalize across multiple languages, opening the door to cross lingual songwriting systems. Lin and He [14] earlier proposed a Joint Sentiment/Topic (JST) model for unsupervised co-discovery of sentiment and theme, and Nandwani and Verma [15] survey the persistent challenges of sarcasm detection, mixed emotion text, and culturally specific affective expressions, all of which remain open issues for any lyric generating system.

### B. AI Driven Lyric Generation

Lyric generation has evolved from n-gram and Markov models in the early 2000s to RNN and LSTM approaches in the mid 2010s, and most recently to LLMs. Lu and Eirinaki [16] demonstrate that GPT-2 fine tuned on lyric corpora produces more fluent and stylistically coherent output than LSTM baselines, while introducing automated scoring across rhyme density, structure, and sentiment balance. Ding et al. [5] propose SongComposer, an LLM that aligns lyrics with melodies at the token level using a unified word and music format. You et al. [17] extend this with S2MILE, which jointly models semantic and structural features of music for lyric coherence. Ma et al. [6] introduce AI-Lyricist, pairing lyrics with MIDI melodies via SeqGAN and post processing to enforce vocabulary constraints. Zhang et al. [18] focus specifically on syllable level alignment between melody and lyrics, while Watanabe and Goto [19] propose a two stage text to lyrics pipeline that uses image based semantic intermediates to reduce paraphrastic plagiarism. Duan et al. [20] introduce a Semantic Dependency Network with unlikelihood training to suppress repetition. Crucially, none of these systems compare multiple state of the art LLMs head to head under controlled experimental conditions, a gap MuseTune directly addresses through its  $3 \times 3$  factorial design.

### C. Chord Generation and Harmony Modeling

Chord generation literature spans rule based, statistical, and deep learning methods. Dervakos et al. [21] propose heuristic scoring grounded in the circle of fifths to evaluate the harmonicity of generated MIDI sequences, providing a rule based baseline against which neural methods can be measured. Li [22] applies LSTMs to multi style chord generation, demonstrating that sequence models can capture long term harmonic dependencies. Ren et al. [23] reformulate chord progressions as a complex network and use random walk algorithms to traverse plausible transitions, an approach conceptually similar to MuseTune’s database retrieval method. Sarmiento [4] explores symbolic guitar tablature generation using Transformer-XL architectures (ShredGP, LooperGP). Zhang and Sun’s MelodyFi [24] generates chords directly from user supplied melodies via key signature analysis, and Chieppa [25] tackles the inverse problem of audio to tab transcription. Despite this breadth, the dominant limitation across this body

of work is that chord generation is rarely conditioned on lyrical sentiment. Chords are typically produced from melody, style tags, or random walks, leaving the emotion to harmony coupling under explored.

### D. Multimodal Creative AI

The intersection of text, audio, and visual generation has produced several notable systems in recent years. Huang et al. [32] describe the AI Song Contest, where teams used modular workflows combining separate models for lyrics, melody, and structure, an architectural philosophy MuseTune shares. The success of these modular approaches over monolithic end to end systems suggests that explicit task decomposition aids both controllability and evaluation. Crothers et al. [30] explore BLOOM-176B for multilingual lyric generation paired with cover art, demonstrating the potential of cross modal creative pipelines.

### E. Evaluation Methodologies for AI Generated Music

A central methodological contribution of MuseTune is its reliance on six peer reviewed, citable metrics rather than ad hoc qualitative demonstrations. Banerjee and Lavie’s METEOR [10] addresses BLEU’s well documented shortcomings for creative text by incorporating stemming and synonymy. Radford et al.’s GPT-2 [3] is widely used as a reference language model for perplexity based fluency scoring. For music specific evaluation, Raffel et al.’s mir\_eval library [26] provides the standard implementation of MIREX chord similarity scoring used across the music information retrieval community. Herremans and Chew’s Spiral Array model [27] enables the quantification of tonal tension over time, capturing dimensions of emotional movement that string similarity metrics cannot. Cuthbert and Ariza’s music21 library [28] supports rule based harmonic validity assessment grounded in classical music theory. Xiong et al. [29] survey AI music evaluation methodologies broadly and recommend the multi metric, mixed method approach that MuseTune adopts. Potash et al. [31] demonstrate the importance of combining computational and human evaluation, particularly for genres like rap where stylistic conventions resist purely automated scoring.

### F. Datasets and Resources

MuseTune’s chord retrieval module draws on two complementary public datasets. **HookTheory** [34] provides crowd sourced chord progressions paired with Spotify audio features (valence, energy, mode), enabling emotion conditioned retrieval. **ChoCo** [8] (Chord Corpus) aggregates over 20,000 annotated chord sequences from multiple sources, distributed in the JAMS format with iReal Pro, Harte, and Roman numeral notations, providing the largest open chord corpus available at the time of writing. For audio synthesis, MuseTune employs the classical **Karplus-Strong** [35] plucked string algorithm for guitar timbre generation, a computationally efficient method requiring no external SoundFonts.

### G. Comparative Positioning

Table I positions MuseTune against ten representative systems from the surveyed literature. While prior work addresses individual components (lyric generation, chord generation, or emotion modeling), MuseTune is, to the best of our knowledge, the first system to (1) integrate all three components in a fully automated pipeline, (2) compare multiple state of the art LLMs against music theoretic baselines through a controlled factorial design, (3) include audio synthesis as part of the demonstrated artifact, and (4) evaluate outputs entirely through citable, domain appropriate metrics rather than human in the loop ratings.

## IV. PROPOSED FRAMEWORK

### IV. PROPOSED FRAMEWORK (MuseTune System Architecture)

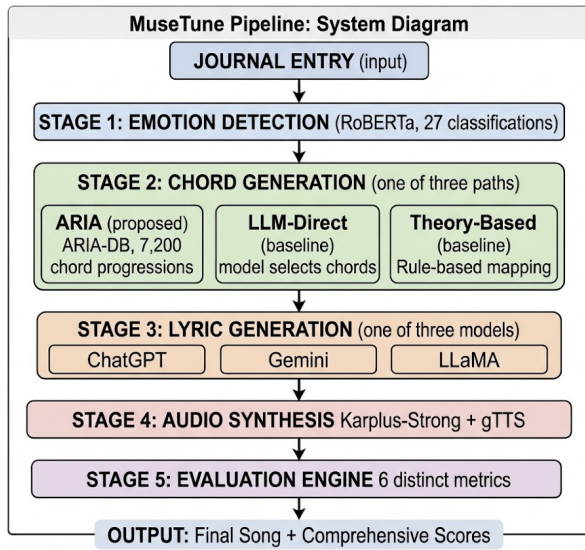


Fig. 1. MuseTune system architecture showing the five sequential pipeline stages.

### A. System Overview

MuseTune is implemented as a modular, fully automated pipeline composed of five sequential stages: (i) Emotion Detection, (ii) Chord Generation, (iii) Lyric Generation, (iv) Audio Synthesis, and (v) Evaluation. The user provides a free text journal entry, and the system returns a complete song artifact containing the detected emotion, a four chord progression, four lines of lyrics, and a playable WAV file.

A defining characteristic of the framework is its support for three interchangeable chord generation methods and three interchangeable lyric generation models, enabling the controlled  $3 \times 3$  factorial evaluation reported in Section V. This modularity allows each component to be replaced or upgraded without re engineering the pipeline. The complete data flow is illustrated in Fig. 1.

### Algorithm 1 ARIA Chord Retrieval

**Require:** Detected GoEmotion  $e$ , chord database  $D$

**Ensure:** Top ranked chord progression  $C^*$

- 1: Map  $e$  to musical bucket  $b$  via  $GoEmotions\_to\_Musical[e]$
- 2: Retrieve filter rules  $r$  from  $EmotionFilters[b]$  (valence, energy, mode bounds)
- 3:  $D_{filtered} \leftarrow$  rows in  $D$  satisfying  $r$
- 4: **if**  $|D_{filtered}| < 5$  **then**
- 5:   Apply relaxed proximity filter (within  $\pm 0.25$  valence)
- 6: **end if**
- 7:  $rank \leftarrow$  frequency count of progressions in  $D_{filtered}$
- 8:  $C^* \leftarrow$  top ranked progression in  $rank$
- 9: **return**  $C^*$

### B. Stage 1: Emotion Detection

The first stage transforms unstructured text into a discrete emotional signal. MuseTune employs the publicly available SamLowe/roberta-base-go\_emotions checkpoint, a RoBERTa transformer fine tuned on the GoEmotions dataset [7]. RoBERTa was selected over alternatives such as BERT or DistilBERT due to its superior performance on emotion recognition benchmarks and its strong handling of informal language patterns common in journal entries.

The model returns a probability distribution over 27 fine grained emotion classes (joy, grief, love, nervousness, admiration, and others). The class with the highest score is selected as the dominant emotion and forwarded to subsequent stages along with its confidence value. This 27 class taxonomy is critical for songwriting expressiveness, as it permits distinctions far finer than the binary positive or negative or six basic emotion schemes that dominate earlier work [11], [13]. For example, the distinction between sadness, grief, disappointment, and remorse, all of which would collapse into a single class in coarser taxonomies, supports meaningful variation in the resulting chord progressions and lyrical content.

### C. Stage 2: Chord Generation (Three Methods)

To rigorously isolate the contribution of music database retrieval, MuseTune implements three alternative chord generation methods that produce a four chord Roman numeral progression from the same emotion input. All three methods accept an emotion label as input and return a list of four Roman numerals, making them directly comparable in the evaluation stage.

1) *ARIA (Proposed)*: This is the system's primary contribution. ARIA queries **ARIA-DB**, a unified corpus of 7,200 chord progressions sourced from HookTheory [34] and ChoCo [8]. The retrieval logic is summarized in Algorithm 1.

For an input emotion, ARIA first maps the GoEmotions label to one of nine coarse musical buckets through a learned correspondence table (for example, joy maps to the joy bucket; amusement, optimism, surprise, and curiosity also map to joy). Each bucket has an associated set of filter rules over valence, energy, and mode. For grief, the filter requires valence  $\leq 0.25$ ,

TABLE I  
COMPARISON OF MUSETUNE WITH RELATED SYSTEMS ACROSS SIX CAPABILITY DIMENSIONS

| System                     | LLM | Lyrics | Emotion | Chords | Audio | n $\geq$ 10 |
|----------------------------|-----|--------|---------|--------|-------|-------------|
| Robinson [2]               | ✓   | ×      | ×       | ✓      | ×     | ×           |
| Ding [5]                   | ✓   | ✓      | ×       | ✓      | ×     | ×           |
| You [17]                   | ✓   | ✓      | ✓       | ×      | ×     | ×           |
| Ma [6]                     | ✓   | ✓      | ✓       | ×      | ×     | ×           |
| Lu [16]                    | ✓   | ✓      | ✓       | ×      | ×     | ×           |
| Sarmiento [4]              | ✓   | ×      | ×       | ✓      | ×     | ×           |
| Li [22]                    | ×   | ×      | ×       | ✓      | ×     | ×           |
| Ren [23]                   | ×   | ×      | ✓       | ✓      | ×     | ×           |
| Zhang [18]                 | ✓   | ✓      | ×       | ×      | ×     | ×           |
| Watanabe [19]              | ✓   | ✓      | ×       | ×      | ×     | ×           |
| <b>MuseTune (proposed)</b> | ✓   | ✓      | ✓       | ✓      | ✓     | ✓           |

energy  $\leq 0.35$ , and mode = 0 (minor key). The corpus is filtered by these rules, and the resulting subset is ranked by frequency, with the most popular progression returned. If the strict filter yields fewer than five candidates, a relaxed proximity filter is applied to maintain reliable output.

2) *LLM-Direct (Baseline)*: The same emotion label is sent directly to Google Gemini 2.5 Flash with a structured prompt requesting a four chord Roman numeral progression. The prompt is engineered to constrain output format and prevent free form text. This baseline isolates the raw harmonic competence of an LLM without database support, testing whether LLMs alone can serve as adequate chord generators.

3) *Theory-Based (Baseline)*: A deterministic rule based mapping converts each emotion to a fixed key and standard progression drawn from classical harmony textbooks [33]. For example, joy maps to C major with I-V-vi-IV; grief maps to A minor with i-VI-III-VII (the Andalusian cadence); fear maps to A minor with i-ii-V-i (diminished approach). This baseline represents traditional music theoretic heuristics encoded as a lookup table.

#### D. Stage 3: Lyric Generation (Three LLMs)

The lyric generation module accepts the detected emotion and the chord progression and prompts a large language model to produce exactly four lines of lyrics. To enable comparative evaluation, three state of the art LLMs are integrated behind a unified interface:

- **OpenAI ChatGPT** (gpt-4o-mini via the OpenAI API)
- **Google Gemini 2.5 Flash** (via the Google Generative AI API)
- **Meta LLaMA 3.3-70B** (via the Groq inference API)

A single shared prompt template enforces consistency across LLMs, requesting four lines that (a) reflect the target emotion, (b) follow loose ABAB or AABB rhyme tendencies, and (c) reference the harmonic mood of the chord progression. The fixed line count ( $n = 4$ ) is essential for the statistical comparability of downstream evaluation metrics across the 90 experiments reported in Section V.

#### E. Stage 4: Audio Synthesis

To convert the symbolic output of stages 1 through 3 into a playable artifact, MuseTune introduces a lightweight audio synthesis stage that operates entirely on the client side without external SoundFonts or MIDI dependencies.

*Guitar synthesis* uses the Karplus-Strong plucked string algorithm [35]. Each chord’s constituent pitches (resolved from the Roman numeral via music21 [28] using the emotion determined key) are synthesized individually with a 0.996 decay coefficient and combined with a 30 millisecond inter string strum delay to emulate the natural attack pattern of a strummed acoustic guitar. The full progression is concatenated into a continuous waveform.

*Vocal synthesis* uses Google’s gTTS engine to produce a spoken rendition of the four lyric lines. While gTTS does not produce sung audio, it provides intelligible vocal narration and serves as a minimal proof of concept for future singing voice integration. Mixing is performed with pydub: the guitar track is attenuated by 6 dB and the vocal track is overlaid with a 1 second pickup offset. The result is exported as a 16 kHz mono WAV file.

#### F. Stage 5: Evaluation Engine

The evaluation engine computes six metrics in parallel for every generated song. Three metrics address the chord progression and three address the lyrics, together covering harmonic correctness, harmonic emotion, harmonic complexity, linguistic fluency, semantic appropriateness, and emotional alignment. The six scores are aggregated into a single weighted Overall Score used for ranking the nine LLM by chord method combinations:

$$\begin{aligned} \text{Overall} = & 0.15 \cdot \text{Harmonic} + 0.15 \cdot \text{ChordSim} \\ & + 0.10 \cdot (1 - \text{Tension}) + 0.20 \cdot \text{Fluency} \\ & + 0.20 \cdot \text{METEOR} + 0.20 \cdot \text{Alignment} \end{aligned}$$

The chord side metrics weight 0.40 in aggregate (Harmonic, ChordSim, and inverted Tension) and the lyric side metrics weight 0.60 (Fluency, METEOR, Alignment), reflecting the relative complexity of lyric evaluation. Tension is inverted

since lower tension generally indicates more emotionally resolved music.

### G. Implementation Details

The full system is implemented in Python 3.12 and runs in Google Colab. Key library dependencies include: transformers (Hugging Face) for RoBERTa and GPT-2; music21 for Roman numeral analysis, key detection, and harmonic validity scoring; mir\_eval for MIREX chord similarity; nltk for METEOR; pydub for audio mixing; gTTS for vocal synthesis; openai, google-genai, and groq for LLM API access; and pandas, numpy, scipy, matplotlib, and seaborn for data handling and visualization. The complete codebase is organized into modular Colab notebook cells, with each cell performing a single conceptual task to maximize readability and reproducibility.

## V. EXPERIMENTAL EVALUATION

### A. Research Questions Addressed

The experimental evaluation directly tests the three research questions stated in Section 1.4. The  $3 \times 3$  factorial design with  $n = 90$  data points supports answering all three questions through aggregate statistics, per emotion analysis, and statistical robustness analysis across journal entries.

### B. Experimental Setup

1) *Environment*: All experiments were conducted in Google Colab with Python 3.12. Compute consisted of a single CPU runtime for most operations and T4 GPU when available for transformer inference. The total compute budget for the 90 experiments was approximately 18 minutes of wall clock time. API costs for OpenAI access totaled approximately USD 0.20 across the full experimental sweep.

#### 2) Datasets:

- **HookTheory**: Crowd sourced chord progressions annotated with Spotify audio features (valence, energy, mode) for approximately 5,000 progressions after deduplication.
- **ChoCo [8]**: 2,200 annotated chord progressions parsed from JAMS format files across four partitions: billboard, rock corpus, and two iReal Pro subsets.
- **GoEmotions [7]**: 58,000 Reddit comments with 27 fine grained emotion labels, used as the training base for our RoBERTa emotion classifier.

The merged ARIA-DB contains 7,200 unique chord progressions after deduplication by (artist, song), with provenance tagging to track source contributions.

#### 3) Evaluation Metrics in Detail: Chord side metrics.

**Harmonic Validity** measures whether a chord progression makes music theoretic sense. It is a weighted composite of three sub scores: a Parse Rate (fraction of chords that music21 can successfully parse), a Resolution Score (fraction of chord transitions that contain a V-I or vii-I cadence), and a Tonic Anchor Score (whether the progression begins and ends on the tonic chord). The final score is  $0.5 \cdot \text{Parse} + 0.3 \cdot \text{Resolution} + 0.2 \cdot \text{Tonic}$ , ranging from 0 to 1.

**Chord Similarity (ChordSim)** uses the mir\_eval library [26] to compute bag of chords similarity between the generated

progression and a reference set of progressions retrieved from ARIA-DB for the same emotion. Higher values indicate the generated progression is closer to human composed reference material.

**Tonal Tension** implements the Spiral Array model of Herremans and Chew [27]. Each Roman numeral chord is mapped to a position on the circle of fifths, and three sub measures are computed: Cloud Diameter (spread of chord positions), Cloud Momentum (rate of harmonic change), and Tensile Strain (mean distance from tonic). The composite tension score is  $0.40 \cdot \text{Diameter} + 0.30 \cdot \text{Momentum} + 0.30 \cdot \text{Strain}$ .

#### Lyric side metrics.

**Fluency** is derived from GPT-2 perplexity [3]. Generated lyrics are passed through the GPT-2 base model, and per token cross entropy loss is computed. The perplexity  $P = e^{\text{loss}}$  is then converted to a 0 to 1 fluency score via  $\text{Fluency} = 1/(1 + P/100)$ . Lower perplexity yields higher fluency.

**METEOR** [10] compares the generated lyrics against an emotion only reference lyric generated by GPT-4o-mini. The reference lyric uses no chord guidance and only the emotion label as input, isolating the contribution of chord conditioning. Higher METEOR indicates greater overlap in vocabulary and structure with the emotion only baseline.

**Emotional Alignment** re detects the emotion from the generated lyrics using the same RoBERTa GoEmotions classifier and scores against the target emotion using a four tier matching scheme: exact match (full credit weighted by detector confidence), related match (same musical bucket, weighted at 0.75 of confidence), weak match (target appears in top 3 detected emotions, weighted at 0.5 of its probability), and mismatch (score of 0).

4) *Composite Score Weights*: The composite Overall Score combines the six metrics using empirically chosen weights (Section 4.6). The chord side weighting (0.40 total) reflects the relatively constrained chord output space (small set of possible progressions), while the lyric side weighting (0.60 total) reflects the higher dimensionality and creative variance of lyric output.

5) *Experimental Design*: We conducted a  $3 \times 3$  factorial design over ten journal entries spanning ten distinct target emotions. For each journal entry, all nine LLM by chord method combinations were evaluated, yielding  $n = 90$  experiments. The ten journal entries were authored to span the GoEmotions space: sadness, joy, anger, fear, excitement, calmness, love, grief, disappointment, and hope.

6) *Test Journal Entries*: Each of the ten journal entries is approximately one to two sentences and reflects a single dominant emotion. Examples include: “I have been sitting in my car in the rain thinking about how much I miss my old friends.” (sadness) and “I just got the promotion I have been working toward for three years and I cannot stop smiling.” (joy). The full set of entries is available in the replication package.



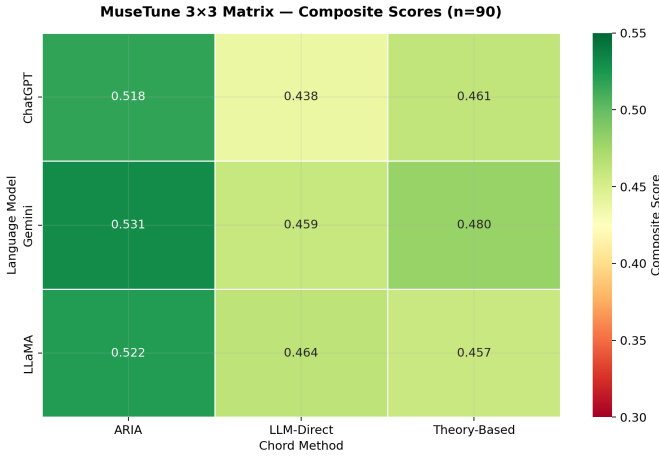


Fig. 2. Composite score heatmap of the 3x3 matrix ( $n = 90$ ). Gemini  $\times$  ARIA achieves the highest score (0.531); ChatGPT  $\times$  LLM-Direct the lowest (0.438). All three ARIA combinations form the strongest column.

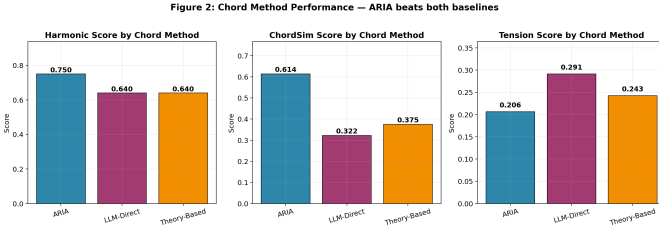


Fig. 3. Chord side metric comparison across the three methods. ARIA dominates Harmonic Validity (0.750) and Chord Similarity (0.614), while maintaining the lowest Tonal Tension (0.206).

## C. Experimental Results

1) *Overall Winners*: Fig. 2 presents the 3x3 composite score heatmap, the headline result of the study. The Gemini  $\times$  ARIA combination achieves the highest overall score of 0.531, followed by LLaMA  $\times$  ARIA at 0.522 and ChatGPT  $\times$  ARIA at 0.518. The worst performing combination is ChatGPT  $\times$  LLM-Direct at 0.438. The entire ARIA column dominates the heatmap regardless of which LLM is paired with it. The gap between the best combination (0.531) and the worst (0.438) is 0.093 absolute composite score points, a substantial difference given the score range of 0 to 1. Notably, all three ARIA configurations cluster within 0.013 of each other (0.518 to 0.531), indicating that the chord method choice exerts a stronger effect on overall quality than the LLM choice.

2) *Chord Method Comparison*: Fig. 3 compares the three chord generation methods across all three chord side metrics. ARIA achieves a Harmonic Validity of 0.750 against 0.640 for both Theory-Based and LLM-Direct, an absolute improvement of 0.110 over both baselines. The most striking gap appears in Chord Similarity: ARIA reaches 0.614, while LLM-Direct collapses to 0.322 and Theory-Based reaches only 0.375. This 0.292 point gap on Chord Similarity is the strongest single result in the study and directly motivates ARIA’s hybrid design: LLM-Direct invents chord combinations rarely seen in

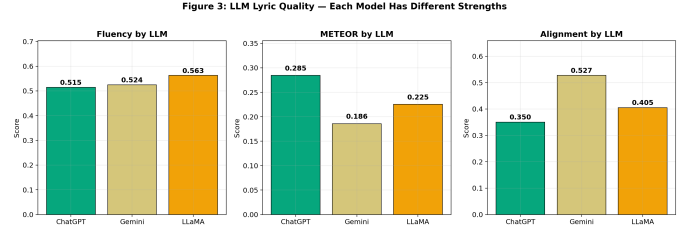


Fig. 4. LLM lyric quality comparison. LLaMA wins on Fluency (0.563), ChatGPT wins on METEOR (0.285), and Gemini wins on Alignment (0.527) by a substantial margin.

human songbooks, while ARIA’s grounding in 7,200 human composed progressions corrects this drift. ARIA also achieves the lowest Tonal Tension (0.206 against 0.291 for LLM-Direct and 0.243 for Theory-Based), indicating more emotionally resolved progressions overall.

This result addresses **RQ1** affirmatively: an emotion to music retrieval algorithm grounded in a curated chord corpus does outperform both pure LLM based and pure music theoretic baselines for emotion conditioned chord generation.

3) *LLM Comparison and Complementary Strengths*: Fig. 4 reveals that no single LLM dominates all lyric side metrics. LLaMA achieves the highest Fluency (0.563), followed closely by Gemini (0.524) and ChatGPT (0.515). ChatGPT leads on METEOR (0.285), with LLaMA (0.225) and Gemini (0.186) trailing. Most strikingly, Gemini dominates Emotional Alignment with a score of 0.527, far ahead of LLaMA (0.405) and ChatGPT (0.350). This suggests three complementary specializations:

- **ChatGPT** excels in emotional vocabulary, producing lyrics that match the lexicon of reference emotional text but at the cost of weaker emotional intensity.
- **LLaMA** excels in natural language flow, producing the most fluent and readable lyrics with moderate emotional alignment.
- **Gemini** excels in emotional expressivity, producing lyrics that re-detect strongly to the target emotion despite weaker vocabulary alignment.

Gemini’s dominance on emotional alignment explains its position as the LLM in the top scoring combination (Gemini  $\times$  ARIA at 0.531). The 0.20 weight on Alignment in the composite score amplifies this advantage. This finding addresses **RQ2** affirmatively: different LLMs exhibit clear complementary strengths and weaknesses when used for emotion conditioned lyric generation.

4) *Statistical Robustness*: Fig. 5 presents box plots of the composite score distribution across ten journal entries for each of the nine combinations, sorted from highest to lowest median. The top three combinations are Gemini  $\times$  ARIA, LLaMA  $\times$  ARIA, and ChatGPT  $\times$  ARIA, all consistently above the grand mean of 0.481. The bottom three combinations are LLaMA  $\times$  Theory-Based, ChatGPT  $\times$  LLM-Direct, and various Theory-Based or LLM-Direct combinations. The observed ranking is stable across journal entries, indicating



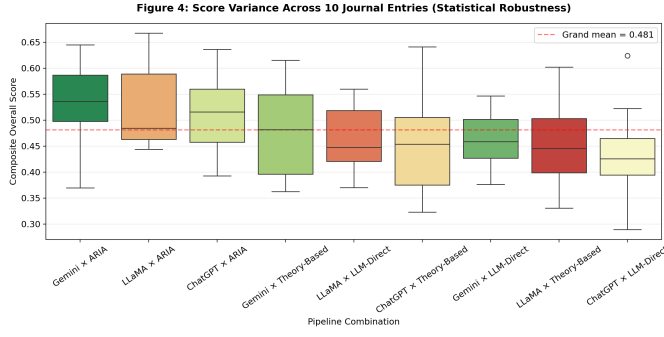


Fig. 5. Composite Overall score distribution across ten journal entries for each of the nine LLM by method combinations. The top three combinations all employ ARIA. The grand mean is 0.481.

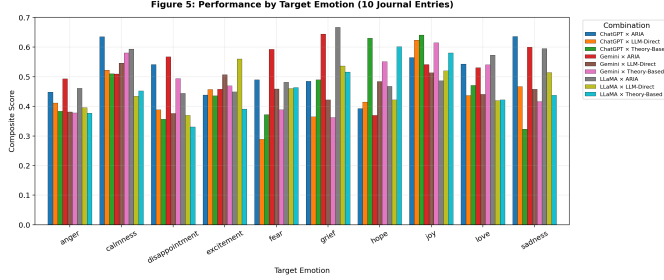


Fig. 6. Composite score by target emotion. ARIA based combinations dominate on calmness, sadness, grief, and fear. Hope and joy show stronger performance from Theory-Based combinations.

that the findings are not driven by outlier inputs. The first three boxes (all ARIA) show similar median heights but moderate variance, suggesting that ARIA’s quality advantage is consistent rather than concentrated in a few favorable inputs.

This robustness analysis directly addresses **RQ3**: a fully automated evaluation framework based on six peer reviewed metrics produces statistically robust rankings across diverse emotional inputs.

5) *Per Emotion Analysis*: Fig. 6 disaggregates the results by target emotion. ARIA performs strongest on contemplative emotions: calmness (ChatGPT  $\times$  ARIA reaches 0.64, the highest single score in the study alongside ChatGPT  $\times$  ARIA on sadness), sadness (ChatGPT  $\times$  ARIA reaches 0.64), grief (LLaMA  $\times$  ARIA reaches 0.67), and fear (Gemini  $\times$  ARIA reaches 0.59). For hope and joy, ChatGPT  $\times$  Theory-Based reaches scores around 0.63 to 0.64, suggesting that textbook progressions in major keys serve positive uplift emotions well. For excitement, LLaMA  $\times$  LLM-Direct reaches around 0.56, indicating that creative chord deviation can support high arousal lyrical content. This finding has important practical implications discussed in Section VI.

6) *Winner vs Loser Metric Profile*: Fig. 7 contrasts the top three and bottom three combinations across five metrics simultaneously. Top combinations (all using ARIA) form large, well rounded polygons that cover substantial area on every metric arm, with Gemini  $\times$  ARIA reaching particularly strong values on Alignment. Bottom combinations form collapsed

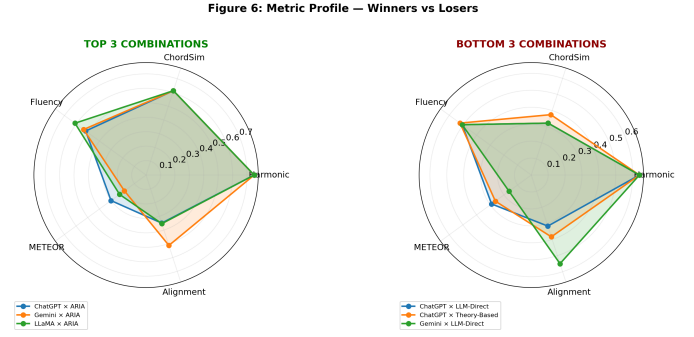


Fig. 7. Radar profile of the top three vs bottom three combinations across five metrics. Top combinations form well rounded polygons; bottom combinations collapse on ChordSim.

TABLE II  
FINAL SUMMARY ACROSS ALL SIX METRICS ( $n = 10$  PER CELL, RANKED BY COMPOSITE)

| LLM     | Method       | Composite    |
|---------|--------------|--------------|
| Gemini  | ARIA         | <b>0.531</b> |
| LLaMA   | ARIA         | 0.522        |
| ChatGPT | ARIA         | 0.518        |
| Gemini  | Theory-Based | 0.480        |
| LLaMA   | LLM-Direct   | 0.464        |
| ChatGPT | Theory-Based | 0.461        |
| Gemini  | LLM-Direct   | 0.459        |
| LLaMA   | Theory-Based | 0.457        |
| ChatGPT | LLM-Direct   | 0.438        |

shapes, with a particularly sharp drop on Chord Similarity. This visualization confirms that winning configurations succeed by performing acceptably across all metrics rather than by maximizing a single one. The implication for AI songwriting system design is clear: balanced performance across all dimensions matters more than excellence in any single one.

7) *Final Summary Table*: Table II presents the final mean scores for each of the nine combinations, ranked by composite score.

## VI. DISCUSSION

### A. Why ARIA’s Hybrid Design Wins

The most consistent finding across all reported figures is that ARIA dominates the entire chord side of the evaluation. The 0.292 point gap on Chord Similarity between ARIA (0.614) and LLM-Direct (0.322) provides strong empirical evidence that LLMs, when used in isolation for chord generation, produce harmonically valid but stylistically unusual progressions that diverge from human composed music. ARIA’s grounding in 7,200 human composed progressions corrects this drift by constraining output to combinations that real musicians actually use. This finding contributes to ongoing debates about whether pretrained LLMs can substitute for domain specific retrieval in symbolic music tasks, and provides a concrete empirical answer: for chord generation specifically, retrieval

over a curated corpus substantially outperforms direct LLM generation.

### B. Chord Method Matters More Than LLM Choice

A central observation of this study is that the three ARIA combinations cluster tightly between 0.518 and 0.531, while combinations sharing the same LLM but different chord methods spread widely (for example, Gemini ranges from 0.459 with LLM-Direct to 0.531 with ARIA, a gap of 0.072). This pattern indicates that the chord method choice exerts a larger effect on overall pipeline quality than the LLM choice. The practical implication is that practitioners building emotion conditioned songwriting systems should prioritize chord retrieval quality first and LLM selection second.

### C. Complementary LLM Strengths

Fig. 4 reveals that ChatGPT, LLaMA, and Gemini exhibit distinct strengths along the three lyric metrics. ChatGPT writes lyrics with the strongest vocabulary alignment to reference emotional text (METEOR 0.285), suggesting access to a rich emotional lexicon. LLaMA produces the most fluent lyrics (Fluency 0.563), suggesting strong language modeling capability. Gemini produces the most emotionally aligned lyrics (Alignment 0.527), suggesting strong emotional expressivity that translates well when the alignment metric is weighted heavily. This complementarity suggests that future songwriting systems may benefit from ensemble or routing approaches, where each LLM contributes to the subtask it handles best.

The dominance of Gemini in the overall ranking despite weaker performance on Fluency and METEOR is largely explained by the 0.20 weight assigned to Alignment in the composite score, combined with Gemini’s substantial lead on that metric (0.527 against LLaMA’s 0.405 and ChatGPT’s 0.350). Practitioners who prioritize different lyric properties may prefer different LLMs.

### D. Emotion Specific Method Selection

The per emotion analysis (Fig. 6) suggests that the optimal chord method may depend on the target emotion. ARIA is strongest on reflective emotions where chord traditions are well established (sadness, grief, calmness, fear). For positive uplift emotions (hope, joy), Theory-Based combinations sometimes match or exceed ARIA, suggesting that textbook progressions in major keys serve these emotions effectively. For high arousal emotions (excitement), LLM-Direct’s creative chord deviation can occasionally outperform retrieval. A future production system could implement an emotion adaptive method selector that routes between methods based on the detected emotion.

### E. The Axis of Awesome Phenomenon

A common observation when running MuseTune is that the system frequently outputs the I-V-vi-IV progression in C major. This is not a bug but a reflection of an empirical regularity of pop music: a significant fraction of contemporary pop songs use I-V-vi-IV in C or G major, a fact popularized

by the comedy group Axis of Awesome in their viral 4 Chords video. MuseTune’s tendency to return these progressions mirrors the distribution of the HookTheory corpus, which itself mirrors the distribution of contemporary pop music. ARIA’s behavior is therefore consistent with real world musicological patterns rather than an algorithmic flaw. Future work could mitigate this by introducing stochastic top N sampling rather than deterministic top 1 retrieval, increasing output diversity at the cost of reduced popularity matching.

### F. Practical Implications for AI Music System Designers

For practitioners building emotion conditioned music systems, the MuseTune evaluation offers three practical lessons. First, do not trust LLMs alone for symbolic music tasks. Use them for tasks they excel at (lyrics, prompts, descriptions) and use retrieval or rule based methods for tasks where domain knowledge matters (chords, melody constraints, structure). Second, evaluate with domain appropriate metrics. Generic NLP metrics such as BLEU miss critical musical dimensions. Third, run factorial experiments. Single combination demos hide the complementary strengths and weaknesses that factorial designs reveal.

### G. Theoretical Implications

The finding that no single LLM dominates all lyric metrics has broader theoretical implications. It suggests that modern LLMs, despite training on overlapping web scale corpora, develop subtly different stylistic and semantic profiles. ChatGPT’s vocabulary advantage may stem from instruction tuning that emphasizes specificity. LLaMA’s fluency advantage may reflect its strong base language modeling. Gemini’s emotional alignment advantage may reflect alignment tuning that calibrates outputs to emotional cues. These differences are masked when systems are evaluated with single aggregate scores, and only emerge through multi metric, multi model factorial studies.

## VII. THREATS TO VALIDITY

We organize threats to validity into three standard categories: internal validity (whether the experimental design measures what it claims to measure), external validity (whether the findings generalize beyond the studied configurations), and construct validity (whether the metrics correspond to the underlying constructs of interest).

### A. Internal Validity Threats

**Emotion to Music Mapping Collapse.** MuseTune collapses GoEmotions’ 27 fine grained classes into nine coarse musical buckets and three keys (C major, A minor, G major). This design ensures statistical comparability across the  $3 \times 3$  matrix but limits output diversity. Two distinct emotions within the same family (for example joy and excitement) often yield identical chord progressions. Future work should explore continuous emotion to music mappings or top N stochastic sampling. We frame this as a deliberate design choice for experimental control rather than a limitation, but note that it does affect the interpretation of per emotion findings.

**Deterministic Top 1 Retrieval.** ARIA’s top 1 retrieval policy means the same emotion always produces the same chord progression. This is desirable for reproducibility but limits creative variance. A more sophisticated retrieval scheme using top K with weighted sampling could improve output diversity without compromising the core evaluation findings.

### B. External Validity Threats

**Limited Corpus Coverage.** ARIA-DB is dominated by pop and rock music from the HookTheory and ChoCo sources. Genres such as jazz, classical, electronic, and traditional non Western music are underrepresented. The system’s notion of harmonic validity is therefore implicitly biased toward Western tonal music. Cross cultural and cross genre evaluation remains future work, particularly for thesis level systems intended for global applicability.

**Fixed Length Lyric Generation.** For statistical comparability across the 90 experiments, all lyrics are constrained to exactly four lines. While this enables fair comparison, it does not test the system on full song structures (verse, chorus, bridge). A separate full song generation mode (18 lines structured as Verse-Chorus-Verse-Chorus-Bridge) has been implemented in the codebase but is not part of the formal evaluation reported here.

**Single Journal Author.** The ten test journal entries were authored by a single individual. While they were written to span the emotional space, they may reflect that author’s particular linguistic patterns. Multi author evaluation would strengthen generalization claims.

### C. Construct Validity Threats

**Reliance on LLM Black Box Components.** The lyric generation step depends on three partially closed source LLMs (ChatGPT, Gemini, LLaMA). Performance reproducibility is therefore subject to API version changes by OpenAI, Google, and Meta or Groq. We document the specific model versions used (gpt-4o-mini, gemini-2.5-flash, llama-3.3-70b-versatile) to maximize reproducibility, but cannot guarantee identical results in perpetuity.

**METEOR Reference Construction.** The METEOR metric requires a reference text. We generate this reference via GPT-4o-mini using emotion only prompting (no chord conditioning). This creates a circular dependency: ChatGPT generates the reference against which all three LLMs are scored, potentially advantaging ChatGPT. We note that ChatGPT does win on METEOR in our results, which is consistent with this circular dependency. Future work should explore human authored reference texts or rotating reference generators.

**Subjective Quality of Audio Output.** The Karplus-Strong plus gTTS audio engine prioritizes openness and reproducibility over fidelity. Production grade vocal singing (for example via Suno, Bark, or DiffSinger) remains outside MuseTune’s research contribution and is treated as orthogonal future work. Audio quality is therefore not evaluated as one of the six metrics, though it is a salient aspect of user experience.

## VIII. CONCLUSION AND FUTURE WORK

### A. Summary of Contributions

This paper presented MuseTune, a fully automated AI song-writing system that converts journal text into a complete song artifact comprising emotion classification, chord progression, lyrics, and audio. The central methodological contribution is ARIA, an Affect-driven Retrieval algorithm that queries a curated 7,200 progression corpus to select chords matching a detected emotion. Across a  $3 \times 3$  factorial study with  $n = 90$  experiments and six citable evaluation metrics, Gemini  $\times$  ARIA achieved the highest composite score (0.531), with all three ARIA combinations occupying the top three positions. The system has been released as a public Colab notebook for use by researchers and practitioners.

### B. Key Findings

The work yields three principal findings: (1) Hybrid LLM and music database approaches consistently outperform pure LLM and pure rule based approaches for emotion conditioned chord generation, with ARIA winning across all three LLMs and producing all three top ranked combinations. (2) Different LLMs exhibit complementary strengths for lyric generation: ChatGPT for vocabulary, LLaMA for fluency, and Gemini for emotional alignment. The composite ranking favors Gemini due to the heavy weighting of Alignment. (3) The chord method choice exerts a larger effect on overall pipeline quality than the LLM choice, indicating that practitioners should prioritize chord retrieval quality first.

### C. Future Research Directions

Several extensions are immediately tractable. First, expanding the emotion to key mapping from three keys to a fine grained per emotion assignment would increase output diversity. Second, top N stochastic sampling within ARIA would reduce repetition across similar inputs while preserving the corpus grounded quality advantage. Third, integration with neural singing voice models (Bark, DiffSinger, RVC) would replace the current narration with actual sung vocals, addressing the largest remaining quality gap. Fourth, a user study comparing MuseTune outputs against commercial systems such as Suno would complement the automated evaluation presented here with human ratings. Fifth, cross genre and cross cultural ARIA-DB extension would address current corpus bias and enable applications beyond Western pop. Sixth, an emotion adaptive method selector that automatically chooses ARIA, LLM-Direct, or Theory-Based based on the detected emotion could improve overall performance by combining the strengths of each method. Seventh, ensemble LLM approaches that route subtasks to the best performing model (ChatGPT for vocabulary heavy lines, Gemini for emotionally intense lines, LLaMA for fluent connecting lines) could exploit the complementary strengths identified in this study.

### D. Broader Impact

MuseTune lowers the technical barrier to emotionally expressive songwriting, with potential applications in music

education, music therapy, and accessibility tools for non musicians. The open source release encourages reproducibility and downstream research. We note potential negative implications including the displacement of human songwriters and the generation of derivative content from copyrighted training data; we address these in the Ethics Statement.

## IX. ETHICS STATEMENT

This research does not involve human subjects, personally identifiable information, or sensitive data, and therefore does not require formal ethics board approval. MuseTune relies exclusively on publicly available, appropriately licensed datasets: HookTheory’s Theorytab database, the ChoCo corpus, and the GoEmotions dataset. All citations of these resources are provided in the references.

Potential ethical considerations include the following. First, generated lyrics may reflect biases present in the training data of the underlying LLMs (ChatGPT, Gemini, LLaMA), including gender, racial, and cultural biases documented in the LLM safety literature. Users are advised to review generated content critically before public release. Second, the system may produce derivative content stylistically similar to copyrighted source material, particularly for chord progressions widely used in popular music. We frame ARIA-DB as a research corpus and the system as a research artifact rather than a commercial product. Third, AI generated songwriting may displace human songwriters in some contexts. We position MuseTune as a creativity support tool rather than a replacement, intended to lower the entry barrier for novices rather than compete with professional artists. The system follows established ethical AI principles including transparency (full open source release), fairness (multi LLM evaluation reduces single vendor bias), and proper attribution (all data sources cited).

## X. CYBERSECURITY STATEMENT

MuseTune does not collect, store, or process personally identifiable information, sensitive personal data, or financial information. The system operates on user supplied journal text in memory only and does not persist user input beyond the current notebook session. API credentials for OpenAI, Google Gemini, and Groq are stored via Google Colab’s encrypted Secrets feature rather than embedded in source code. For any future public deployment, standard web security practices including HTTPS, rate limiting, and access controls should be enforced. User generated journal text could in principle contain sensitive personal disclosures; users should be informed that LLM API calls send this text to third party providers (OpenAI, Google, Groq) subject to those providers’ privacy policies.

## XI. REPLICATION PACKAGE

All code, datasets, results, and figures used in this paper are publicly available at:

<https://github.com/arpitsoni0/MuseTune>

The repository includes: (a) the full Colab notebook with modular reproducible cells covering all stages from data

loading through evaluation and visualization; (b) the ARIA-DB construction scripts and the merged 7,200 entry CSV with provenance tagging; (c) the raw  $n = 90$  experiment results CSV including all individual metric scores per (LLM, method, journal) triple; (d) the six thesis figures (heatmap, chord method bars, LLM bars, variance box plot, per emotion bars, and radar) along with the matplotlib generation scripts; (e) the final summary table CSV used in Section 5; and (f) a comprehensive README describing installation, API key setup, and step by step reproduction of all reported numbers and figures.

## ACKNOWLEDGMENTS

The author thanks Dr. Hussain Ahmed of the Adelaide University for supervisory guidance throughout this research. The author also acknowledges the HookTheory team, the ChoCo project contributors, Google Research for the GoEmotions dataset, and the open source maintainers of music21, mir\_eval, and transformers whose tools made this work possible.

## REFERENCES

- [1] Y. Zhao et al., “AI-enabled text-to-music generation: A comprehensive review,” Preprints, Feb. 2025. [Online]. Available: <https://doi.org/10.20944/preprints202502.1791.v1>
- [2] C. Robinson, A. Yang, and T. Liu, “LLM music creation and recommendation applications,” B.S. thesis, Dept. of Electrical and Computer Engineering, Massachusetts Institute of Technology, Cambridge, MA, USA, Jun. 2025.
- [3] A. Radford, J. Wu, R. Child, D. Luan, D. Amodei, and I. Sutskever, “Language models are unsupervised multitask learners,” OpenAI Technical Report, 2019.
- [4] P. P. Sarmiento, “Guitar tablature generation with deep learning,” Ph.D. dissertation, Queen Mary Univ. of London, 2024.
- [5] S. Ding et al., “SongComposer: A large language model for lyric and melody generation in song composition,” in Proc. ACL, 2024.
- [6] X. Ma et al., “AI-Lyricist: Generating music and vocabulary constrained lyrics,” in Proc. ACM Multimedia, 2024.
- [7] D. Demszky, D. Movshovitz-Attias, J. Ko, A. Cowen, G. Nemade, and S. Ravi, “GoEmotions: A dataset of fine-grained emotions,” in Proc. ACL, 2020, pp. 4040–4054.
- [8] J. de Berardinis, A. Meroño-Peñuela, A. Poltronieri, and V. Presutti, “ChoCo: A chord corpus and a data transformation workflow for musical harmony knowledge graphs,” Sci. Data, vol. 10, no. 641, 2023.
- [9] H. Gill, D. T. Lee, and N. Marwell, “Deep learning in musical lyric generation: An LSTM-based approach,” Yale Univ., USA.
- [10] S. Banerjee and A. Lavie, “METEOR: An automatic metric for MT evaluation with improved correlation with human judgments,” in Proc. ACL Workshop, 2005, pp. 65–72.
- [11] W. Medhat, A. Hassan, and H. Korashy, “Sentiment analysis algorithms and applications: A survey,” Ain Shams Eng. J., vol. 5, no. 4, pp. 1093–1113, 2014.
- [12] A. Ara and V. Rekha, “Emotion classification of music lyrics using LSTM,” in Proc. ICMCSI, 2024.
- [13] M. S. U. Miah et al., “A multimodal approach to cross-lingual sentiment analysis with ensemble of transformer and LLM,” Sci. Rep., vol. 14, no. 9603, 2024.
- [14] C. Lin and Y. He, “Joint sentiment/topic model for sentiment analysis,” in Proc. CIKM, 2009, pp. 375–384.
- [15] P. Nandwani and R. Verma, “A review on sentiment and emotion detection from text,” Soc. Netw. Anal. Min., vol. 11, no. 81, 2021.
- [16] J. Lu and M. Eirinaki, “Can a machine win a Grammy? An evaluation of AI-generated song lyrics,” in Proc. IEEE Big Data, 2021, pp. 4897–4906.
- [17] M. You, F. Zhang, S. Zhang, and L. Xu, “S2MILE: Semantic-and-structure-aware music-driven lyric generation,” in Proc. AAAI, 2024.
- [18] Z. Zhang, Y. Yu, and A. Takasu, “Controllable syllable-level lyrics generation from melody,” IEEE Trans. Multimedia, vol. 26, pp. 11084–11096, 2024.

- [19] K. Watanabe and M. Goto, "Text-to-lyrics generation with image-based semantics," ISMIR, 2023.
- [20] W. Duan, Y. Yu, and K. Oyama, "Semantic dependency network for lyrics generation," Neural Comput. Appl., 2023.
- [21] E. Dervakos, G. Filandrianos, and G. Stamou, "Heuristics for evaluation of AI-generated music," IEEE, 2023.
- [22] F. Li, "Chord-based music generation using long short-term memory neural networks," J. Supercomputing, Oct. 2023.
- [23] T. Ren et al., "Guitar chord-generating algorithm based on complex network," Physica A, vol. 450, pp. 302–314, 2016.
- [24] J. Zhang and Y. Sun, "An intelligent system to assist piano composition and chords generation," CS & IT.
- [25] S. Chieppa, "Automatic guitar transcription using deep neural networks," MSc thesis, Univ. of Coimbra, 2024.
- [26] C. Raffel, B. McFee, E. J. Humphrey, J. Salamon, O. Nieto, D. Liang, and D. P. W. Ellis, "mir\_eval: A transparent implementation of common MIR metrics," in Proc. ISMIR, 2014.
- [27] D. Herremans and E. Chew, "Tension ribbons: Quantifying and visualising tonal tension," in Proc. TENOR, 2017.
- [28] M. S. Cuthbert and C. Ariza, "music21: A toolkit for computer-aided musicology and symbolic music data," in Proc. ISMIR, 2010, pp. 637–642.
- [29] Z. Xiong, W. Wang, J. Yu, Y. Lin, and Z. Wang, "A comprehensive survey for evaluation methodologies of AI-generated music," arXiv preprint arXiv:2308.13736, 2023.
- [30] E. Crothers, H. Viktor, and N. Japkowicz, "BLOOM: Creativity and affinity in artificial lyrics and art," arXiv preprint arXiv:2301.05402, 2023.
- [31] P. Potash, A. Romanov, and A. Rumshisky, "Evaluating creative language generation," in Proc. EMNLP, 2018.
- [32] C.-Z. A. Huang et al., "AI Song Contest: Human-AI co-creation in songwriting," in Proc. ISMIR, 2020.
- [33] W. Piston, *Harmony*. W. W. Norton, 1941.
- [34] HookTheory LLC, "Theorytab database," 2024. [Online]. Available: <https://www.hooktheory.com/theorytab>
- [35] K. Karplus and A. Strong, "Digital synthesis of plucked-string and drum timbres," Comput. Music J., vol. 7, no. 2, pp. 43–55, 1983.